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$$\begin{aligned}
 & \int_0^1 \ln x dx \\
 &= \lim_{a \rightarrow 0+} \int_a^1 \ln x dx \\
 & \left\{ \begin{array}{l} u = \ln x \\ dv = dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = \frac{1}{x} dx \\ v = x \end{array} \right. \\
 &= \lim_{a \rightarrow 0+} \left(x \ln x - \int x \cdot \frac{1}{x} dx \right)_a^1 \\
 &= \lim_{a \rightarrow 0+} \left(x \ln x - \int 1 dx \right)_a^1 \\
 &= \lim_{a \rightarrow 0+} (x \ln x - x)_a^1 \\
 &= \lim_{a \rightarrow 0+} [(1 \cdot \ln(1) - 1) - (a \ln(a) - a)] \\
 &= \lim_{a \rightarrow 0+} (-1 - a \ln(a) + a) \\
 & \left\{ \begin{array}{l} \lim_{a \rightarrow 0+} a \ln(a) = 0 \cdot (-\infty) \text{ onbepaald} \\ \Rightarrow \frac{\ln(a)}{\frac{1}{a}} = \frac{-\infty}{+\infty} \stackrel{\text{L'Hôpital}}{=} \lim_{a \rightarrow 0+} \frac{\frac{1}{a}}{-\frac{1}{a^2}} = \lim_{a \rightarrow 0+} \frac{1}{a} \cdot (-a^2) = \lim_{a \rightarrow 0+} -a = 0 \end{array} \right. \\
 &= -1 - 0 + 0 \\
 &= -1 \\
 & \int_0^1 \ln x dx = -1
 \end{aligned}$$

References